

GATE Questions: Stack, Queue and Linked List

1. What is a stack?
2. Which principle does a stack follow?
3. What is the time complexity of the Push operation?
4. What is the time complexity of the Pop operation?
5. What is stack overflow?
6. What is stack underflow?
7. How is a stack implemented using an array?
8. How is a stack implemented using a linked list?
9. Why is a stack used in recursion?
10. How is a stack used in function calls?
11. How do you convert an infix expression to postfix using a stack?
12. How do you evaluate a postfix expression using a stack?
13. How is a stack used for balanced parentheses checking?
14. What is the difference between recursion and an explicit stack?
15. How can two stacks be implemented in a single array?
16. Design a stack that supports getMin() in $O(1)$ time.
17. What are the applications of a stack in computer science?
18. What is a queue?
19. Which principle does a queue follow?
20. What is the time complexity of Enqueue and Dequeue operations?
21. What is queue overflow?
22. What is queue underflow?
23. What is a circular queue?
24. Why is a circular queue preferred over a linear queue?
25. What is a deque?
26. What is a priority queue?
27. What is the difference between a queue and a deque?
28. How is a queue implemented using an array?
29. How is a queue implemented using a linked list?
30. What are the applications of a queue?
31. Why is a queue used in BFS?
32. How can a queue be implemented using two stacks?
33. What is the purpose of Front and Rear pointers?
34. What is the time complexity of Peek?
35. What is a linked list?

36. What is the difference between an array and a linked list?
37. What is a singly linked list?
38. What is a doubly linked list?
39. What is a circular linked list?
40. What is the time complexity of insertion at the beginning?
41. What is the time complexity of deletion at the beginning?
42. What is the time complexity of searching?
43. How do you reverse a singly linked list?
44. How do you find the middle node?
45. How do you detect a loop?
46. What is Floyd's Cycle Detection Algorithm?
47. How do you merge two sorted linked lists?
48. What are the advantages of a linked list over an array?
49. What are the disadvantages of a linked list?
50. How can stacks and queues be implemented using linked lists?